The University of Texas at Austin

Multimodal integration reveals complementary imaging and genetic predictors of coronary artery disease risk

Oden Institute for Computational Engineering and Sciences

Devansh Pandey[1], Liaoyi Xu[1], Eucharist Kun[1], Chenfei Li[1], Joyce Y. Wang[1], Alaa Melek[2], Julie C. DiCarlo[3,5], Edward Castillo[2], Jagat Narula[5], Charles A. Taylor[3,5], Vagheesh M. Narasimhan[1,3,6,†]

[1]Department of Integrative Biology, The University of Texas at Austin · [2]Department of Biomedical Engineering, The University of Texas at Austin · [3]Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin · [4]University of Texas Health Sciences Center, Houston · [5]Department of Internal Medicine, Dell Medical School, The University of Texas at Austin · [6]Department of Statistics and Data Science, The University of Texas at Austin · [†]Corresponding author · Presenting author in bold

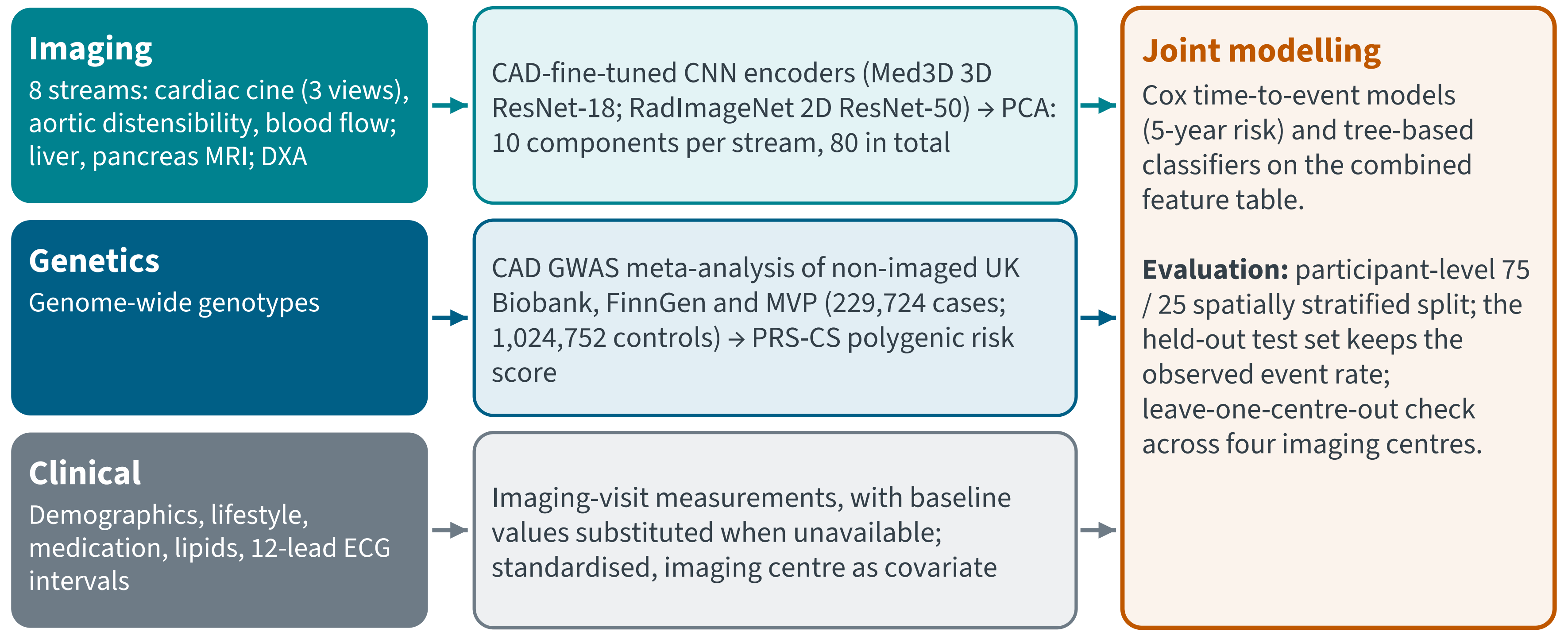
Take-home Cardiac imaging and polygenic risk capture complementary information about future CAD, improving risk stratification beyond clinical variables in UK Biobank.

Study design and imaging features

Clinical risk scores use a small set of traditional variables. We asked which imaging modalities add predictive information for incident coronary artery disease (CAD) beyond clinical, electrocardiogram (ECG) and genetic risk, and how imaging-derived and polygenic risk relate in the same people.

42,351	677	4.6 y	75 / 25
participants, UK Biobank imaging cohort	incident CAD cases (1.6% observed event proportion)	median follow-up; 5-year risk horizon	training 31,754 (499 events) / held-out test 10,597 (178 events)

Multimodal workflow



Imaging
8 streams: cardiac cine (3 views), aortic distensibility, blood flow; liver, pancreas MRI; DXA

CAD-fine-tuned CNN encoders (Med3D 3D ResNet-18; RadImageNet 2D ResNet-50) → PCA: 10 components per stream, 80 in total

Genetics
Genome-wide genotypes

CAD GWAS meta-analysis of non-imaged UK Biobank, FinnGen and MVP (229,724 cases; 1,024,752 controls) → PRS-CS polygenic risk score

Clinical
Demographics, lifestyle, medication, lipids, 12-lead ECG intervals

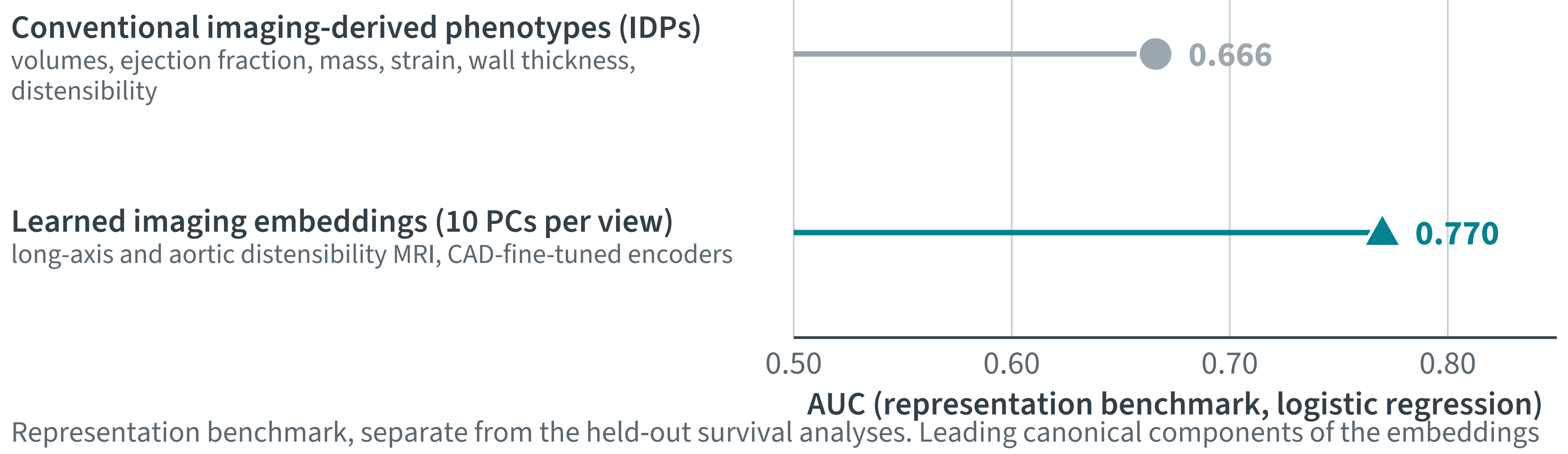
Imaging-visit measurements, with baseline values substituted when unavailable; standardised, imaging centre as covariate

Joint modelling
Cox time-to-event models (5-year risk) and tree-based classifiers on the combined feature table.

Evaluation: participant-level 75 / 25 spatially stratified split; the held-out test set keeps the observed event rate; leave-one-centre-out check across four imaging centres.

Outcome: clinically recognised CAD (myocardial infarction, angina or ischaemic heart disease, coronary revascularisation) from linked records; first event after the imaging visit, prevalent CAD excluded. Cardiac MRI does not image coronary plaque. Encoders were fine-tuned on balanced case–control sets from the training partition only; all risk analyses keep the observed event rate. Genetically defined White British participants.

Learned imaging representations outperform conventional phenotypes

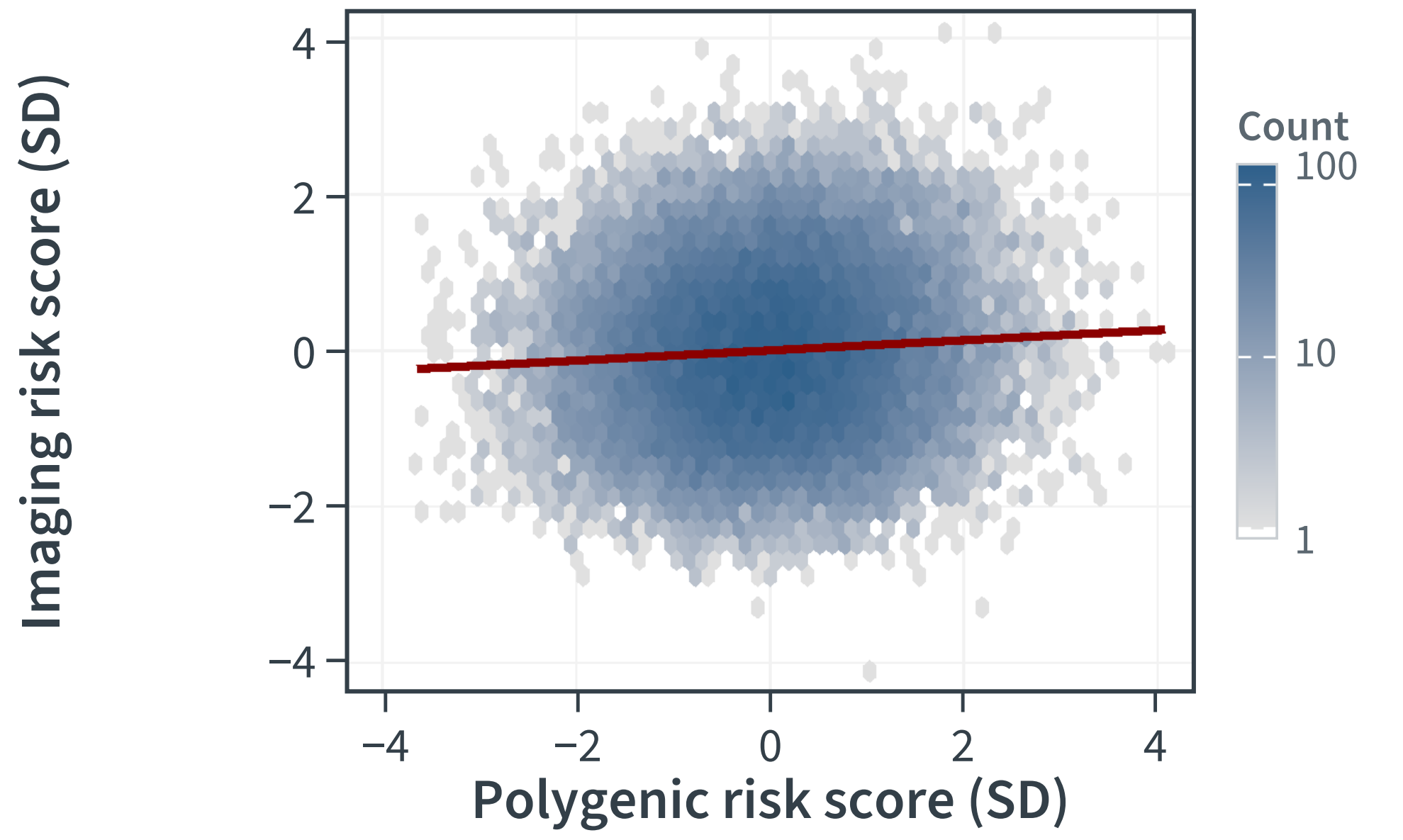


Representation benchmark, separate from the held-out survival analyses. Leading canonical components of the embeddings align with stroke volume, ejection fraction and myocardial mass (canonical correlations 0.79–0.84).

Imaging and genetics capture complementary risk

An imaging risk score (IRS; ridge-penalised Cox model on the 80 imaging principal components, five-fold cross-fitted) and the PRS both predict CAD, yet they barely overlap.

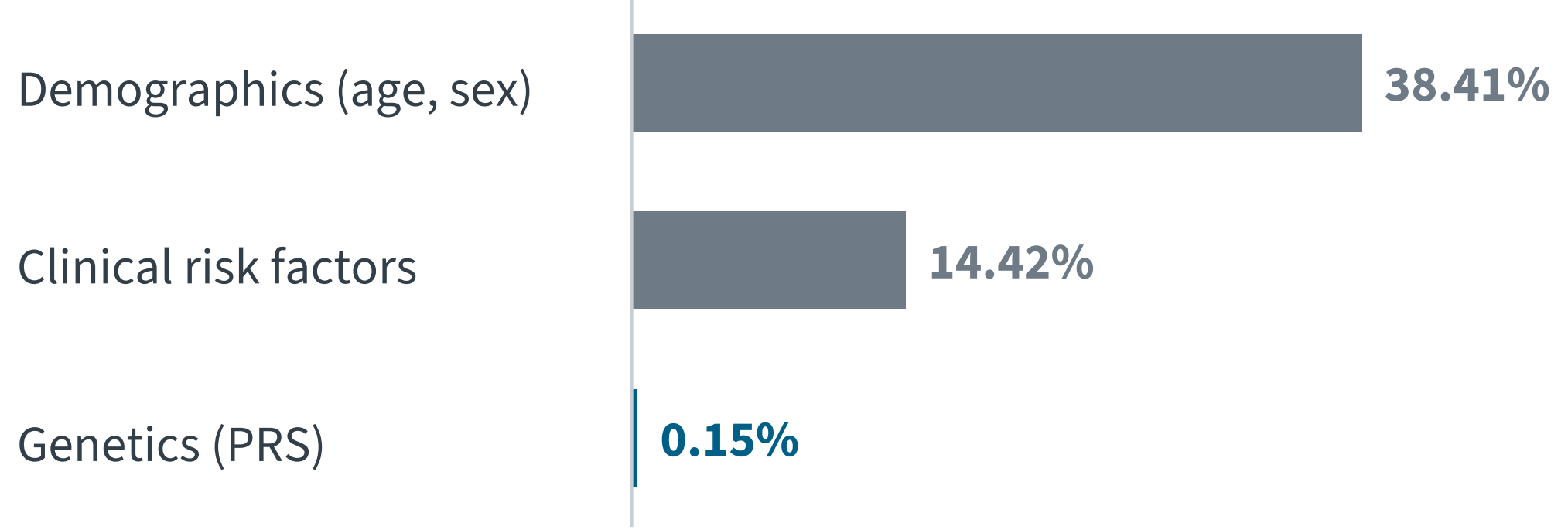
Weak PRS–imaging risk correlation



PRS–imaging risk correlation: r = 0.065; shared variance = 0.4%; n = 29,075.

Low overlap supports complementarity; it does not establish biological independence or show that imaging risk is wholly acquired.

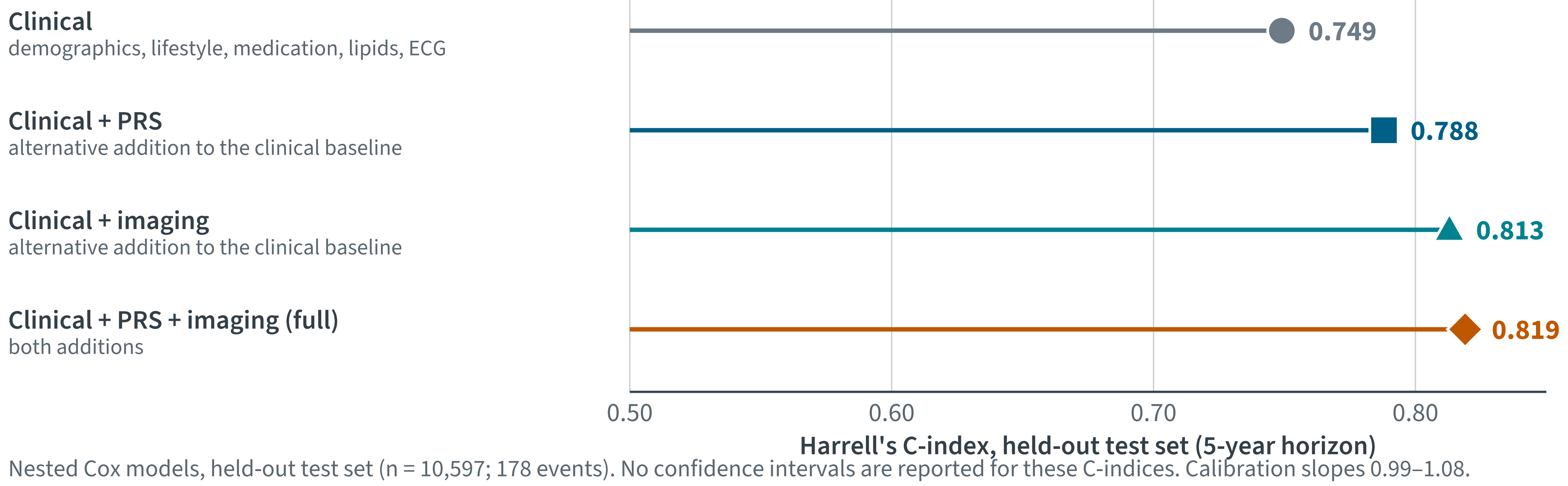
Variance of the imaging risk score explained (partial R^2)



Which feature groups carry non-redundant information? (leave-one-group-out)

Largest loss when removed	Little loss after clinical adjustment
Long-axis cardiac MRI	Liver MRI
Medication history	Pancreas MRI
Polygenic risk score	Whole-body DXA
Aortic distensibility MRI	Blood-flow MRI

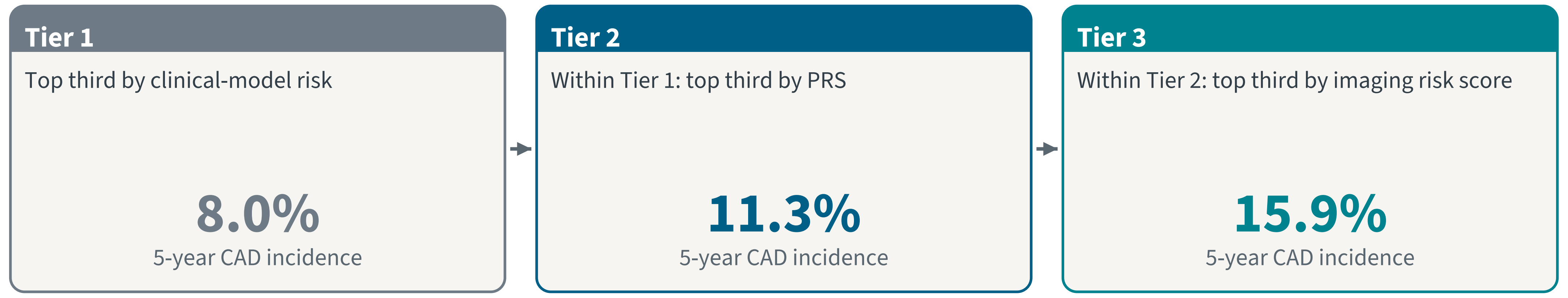
Held-out discrimination: full model 0.819 vs clinical 0.749 (absolute difference 0.070)



Nested Cox models, held-out test set (n = 10,597; 178 events). No confidence intervals are reported for these C-indices. Calibration slopes 0.99–1.08.

Combined models refine CAD risk

Sequential stratification of the held-out set: each tier keeps the top third of the previous tier, with the cut-point recalculated within that tier.



Tier 1
Top third by clinical-model risk
8.0%
5-year CAD incidence

Tier 2
Within Tier 1: top third by PRS
11.3%
5-year CAD incidence

Tier 3
Within Tier 2: top third by imaging risk score
15.9%
5-year CAD incidence

Within Tier 3, the top within-tier quintile of both PRS and imaging risk reached 36.8% five-year incidence; stratum size and confidence interval are not reported. For context, the crude event proportion in the full cohort is 1.6%.

Leave-one-centre-out evaluation within UK Biobank (not external validation)

Held-out centre	Test cases / controls	AUC	C-index
Reading	71 / 7,607	0.805	0.762
Cheadle	409 / 21,942	0.822	0.748
Bristol	82 / 3,307	0.785	0.751
Newcastle	115 / 8,740	0.795	0.732

Site-analysis models trained on three centres and tested on the fourth. Different feature set and validation design from the main held-out analysis.

Calibration, reclassification and accuracy (held-out)

0.634	5.5% → 9.5%	0.99–1.08
continuous NRI, full vs clinical (95% CI 0.534–0.734)	index of prediction accuracy, clinical → full	calibration slopes, all nested models

Full-model net benefit was highest at 5–30% threshold probabilities. Continuous NRI is not the share of people correctly reclassified.

Conclusion

Selected cardiac imaging (long-axis cine and aortic distensibility MRI) and genome-wide polygenic risk add partially non-overlapping information to clinical variables, improving discrimination and risk enrichment, with calibration slopes near 1.

Limitations

- Selected, complete-case, White British UK Biobank sample; absolute risks may not transport to other settings.
- No fully external validation; site robustness is within one cohort.
- No coronary calcium or CT-angiography benchmark; cardiac MRI does not image plaque.
- Observational prediction: clinical utility, test ordering and cost-effectiveness remain untested.
- Extreme-risk strata are small and their precision is not reported.

Data and code

UK Biobank data are available to approved researchers through application (this work: application 65439). Analysis and figure code: github.com/Devanshpandey/Multimodal-integration-for-coronary-artery-disease-risk-prediction (code only; no participant data or model weights). Genome-wide polygenic risk score; no individual gene-level discovery claims.

Preprint (earlier version)

medRxiv doi:10.1101/2025.10.09.25337677

Acknowledgements

Computing: Texas Advanced Computing Center (Director's Discretionary Fund). D.P.: Outstanding Graduate Research Fellowship, The Graduate School, UT Austin. V.M.N.: Allen Discovery Center program (Paul G. Allen Frontiers Group, Paul G. Allen Family Foundation) and Good Systems Fellowship for Ethical AI, UT Austin.

Disclosures

C.A.T. founded HeartFlow, Inc. and serves on the boards of HeartFlow, Inc. and Ebenbuild GmbH. The other authors declare no competing interests.